

Heimdall

- Security Operations & Intrusion Detection Platform
- See Everything. Protect Everyone.

Project Overview

- Modular cybersecurity platform
- Log analysis
- IDS
- Infrastructure monitoring

Project Vision

- Real-time visibility
- Threat detection
- System health monitoring

Core Components

- Log Ingestion
- IDS
- Alerting
- Monitoring
- Correlation
- Dashboard

Log Ingestion

- Reads raw logs
- Converts into structured data

IDS

- Detects suspicious patterns
- Brute force
- IP anomalies

Alerting Engine

- Generates alerts
- Severity levels

Infrastructure Monitoring

- Tracks machines and services
- Agentless + Agent-based

Agentless Monitoring

- Polling via TCP/HTTP
- No installation required

Agent-Based Monitoring

- Heartbeat signals
- Metrics collection

Monitoring Core

- Processes monitoring data
- UP/DOWN/DEGRADED

Correlation Layer

- Combines logs + monitoring
- Detects complex issues

SOC Dashboard

- Visualizes alerts
- System status
- Charts

Alert Types

- Security
- Infrastructure
- Performance
- Correlated

Detection Rules

- Failed logins
- Server down
- High latency

Architecture

- Data → Processing → Alerts → Dashboard

What You Will Build

- Backend
- Monitoring
- Detection
- Dashboard

What You Will Learn

- Backend dev
- System design
- Security basics

Why It Matters

- Real SOC simulation
- Multi-domain project

Final Goal

- What is happening?
- Is it dangerous?
- Is system alive?